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Fischer−Tropsch Synthesis to Lower Olefins over Potassium-Promoted Reduced Graphene Oxide Supported Iron Catalysts

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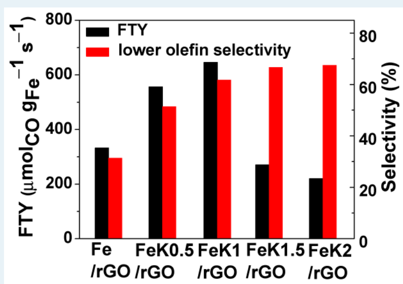
Supporting Information

ABSTRACT: Fischer−Tropsch synthesis to lower olefins (FTO) opens up a compact and economical way to the production of lower olefin directly from syngas (CO and H_2) derived from natural gas, coal, or renewable biomass. The present work is dedicated to a systematic study on the effect of K in the reduced graphene oxide (rGO) supported iron catalysts on the catalytic performance in FTO. It is revealed that the activity, expressed as moles of CO converted to hydrocarbons per gram Fe per second (iron time yield to hydrocarbons, termed as FTY), increased first with the content of K, passed through a maximum at 646 $\mu mol_{CO}\ g_{Fe}^{-1}\ s^{-1}$ over the FeK1/rGO catalyst, and then decreased at higher K contents. Unlike the evolution of the activity, the selectivity to lower olefins increased steadily with K, giving the highest selectivity to lower olefins of 68% and an olefin/paraffin (O/P) ratio of 11 in the C_2−C_4 hydrocarbons over the FeK2/rGO catalyst. The volcanic evolution of the activity is attributed to the interplay among the positive effect of K on the formation of Hägg carbide, the active phase for FTO, and the negative roles of K in increasing the size of Hägg carbide at high content and blocking the active phase by K-induced carbon deposition. The monotonic increase in the selectivity to lower olefins is ascribed to the improved chain-growth ability and surface CO/H_2 ratio in the presence of K, which favorably suppressed the unwanted CH_4 production and secondary hydrogenation of lower olefins.



KEYWORDS: *iron catalysts, potassium promoter, reduced graphene oxide, Fischer−Tropsch synthesis, lower olefins*

INTRODUCTION

Lower olefins, i.e., ethylene, propylene, and butylenes, are important intermediates in chemical industries for the production of polymers, solvents, drugs, cosmetics, detergents, and so on.[1] Conventionally, lower olefins are manufactured on a massive scale from unsustainable crude oil.[2] Taking into account concerns regarding resources and the environment, strategies that start with syngas derived from natural gas, coal, or renewable biomass for the production of lower olefins are more desirable. In general, these syngas-based strategies can be categorized into indirect processes involving methanol to olefins (MTO) and dimethyl ether to olefins (DMTO) that require the synthesis of intermediate products such as methanol and dimethyl ether, respectively, and a direct process on the basis of FTO that is obviously environmentally more interesting and economically more profitable.[3]

In FTO, iron occupies an indispensable status not only due to its low cost and high water-gas shift activity[4] but also for its high selectivity to lower olefins with low CH_4 productivity at the high reaction temperatures required by FTO.[3,5] Recently, de Jong and co-workers reported that the weak interactive α-Al_2O_3-supported iron catalyst modified with Na and S could afford a high selectivity to lower olefins of 53% and a high O/P ratio of 8.8 with an FTY of 13.5 $\mu mol_{CO}\ g_{Fe}^{-1}\ s^{-1}$ at 613 K, 20 bar, and a H_2/CO ratio of 1.[1] Inspired by that work, Zhou et al. prepared hierarchically structured α-Al_2O_3-supported iron catalysts modified with S for FTO.[6] Despite the the fact that high selectivity to lower olefins of 68% was obtained at 623 K, 1 bar, and a H_2/CO ratio of 1, the FTY was only 5.9 $\mu mol_{CO}\ g_{Fe}^{-1}\ s^{-1}$ at a CO conversion of 0.9%.

Regarding the moderate metal−support interaction,[7,8] carbon materials supported iron catalysts are promising alternatives to the Fe/α-Al_2O_3 catalysts for FTO. In the same work cited above, de Jong and co-workers found that a carbon nanofiber (CNF) supported iron catalyst modified with Na and S was only slightly less selective than the α-Al_2O_3-supported counterpart.[1] Muhler and co-workers reported a high FTY of 595.5 $\mu mol_{CO}\ g_{Fe}^{-1}\ s^{-1}$ on N-functionalized multiwalled carbon

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nanotube (CNT) supported iron nanoparticles (NPs).[9] However, the percentage of olefins in the C_2-C_6 fraction was not appealing. Xu et al. reported that, on CNT-supported Fe−Mn NPs, the selectivity to lower olefins was 31.5% whereas that to C_{5+} was as high as 53.8%.[10] On Mn and K-coated CNT-supported iron catalysts, high selectivity to lower olefins of 50% and O/P ratio of 6.6 were obtained at a CO conversion of 22.7%.[11] Recently, Chen et al. employed N-doped graphene as the support for iron NPs (Fe/NG), on which the selectivity to lower olefins was close to 50%, implying that graphene is a promising support for iron in FTO.[12] However, these carbon material supported iron catalysts, similar to the case for many other supported iron catalysts, are not able to afford concurrently high activity and high selectivity to lower olefins, thus resulting in low iron time yield to lower olefins (FTY_{ole}, moles of CO converted to lower olefins per gram of Fe per second).[1]

As a newly emerging carbon material, graphene is regarded as an ideal catalyst support owing to its characteristics such as large surface area, unique two-dimensional (2D) structure, excellent electrical and thermal conductivity, and ease of modification.[13] Graphene that is reduced from graphene oxide (GO) is usually termed as rGO.[13] In this work, we prepared K-promoted rGO-supported iron NP catalysts (FeK/rGO) for FTO with the aim of disclosing the effects of K imposed on the microstructure and catalytic performance, which have not been exploited on graphene-supported iron catalysts. The main merits of rGO include large surface area, abundant oxygen-containing groups, and defects,[7,13] which are expected to be beneficial for anchoring and dispersing the iron NPs.[14] The unique 2D structure of rGO may facilitate the desorption of the reaction intermediate.[15] In FTO, lower olefins are the reaction intermediates, which tend to be readsorbed and then undergo secondary reactions;[3] thus, rGO may afford a higher selectivity to lower olefins.[16] K is effective in enhancing the FTS activity at low concentrations.[17,18] More importantly, it is also able to improve the selectivity to lower olefins.[19,20] In this connection, the FeK/rGO catalysts show promise to exhibit both high activity and high selectivity to lower olefins in FTO. The excellent thermal conductivity of graphene may additionally endow the FeK/rGO catalysts excellent stability in FTO operating at high temperature.

■ EXPERIMENTAL SECTION

Catalyst Preparation. GO (XF Nano Materials Technol.) was used as received, which contained 1.8 wt % of sulfur, 0.6 wt % of chlorine, and 0.16 wt % of potassium determined on a Bruker AXS S4 Explorer X-ray fluorescence (XRF) scattering spectrometer. Other chemicals were of analytical grade (AR) and were obtained from Sinopharm Chemical Reagent. The gases were obtained from Shanghai Youjiali.

For the preparation of the Fe/rGO catalyst, 0.50 g of GO was dispersed in 100 mL of deionized water by ultrasonication for 2 h to yield a yellowish brown suspension, to which 0.50 g of $Fe(acac)_3$ dissolved in 25 mL of ethanol was added with stirring. After the suspension was stirred at room temperature for 4 h to homogeneity, the mixture was heated to 363 K, and 10 mL of 85% hydrazine hydrate aqueous solution acting as both the reducing agent for GO and the precipitant for ferric ions was added dropwise with vigorous stirring. A black precipitate appeared in a few minutes, indicating the fast reduction of GO to rGO. Meanwhile, the pH value increased from 1.5 to 8.5, possibly due to the occurrence of $N_2H_4 + H_2O = N_2H_5^+ + OH^-$, which led to the concomitant precipitation of ferric ions as $Fe(OH)_3$, as $Fe(acac)_3$ is not stable in a basic environment. The mixture was then refluxed at 363 K for 12 h. The resulting black solid was collected by filtration, washed repeatedly with deionized water, and dried at 333 K for 8 h. Since hydrazine hydrate was used to reduce GO, the resulting Fe/rGO contained a small amount of N (1.5 wt %) as determined on the VARIO EL3 instrument (ELEMENTAR), which is consistent with the finding of Ruoff and co-workers.[21]

The FeK/rGO catalysts were prepared by impregnating Fe/rGO with K_2CO_3. After the washing step in the preparation of Fe/rGO, the Fe/rGO cake (dry weight 0.40 g) was reslurried in 200 mL of deionized water and sonicated for 30 min. Then, a calculated volume of the K_2CO_3 aqueous solution was added to give the prescribed K content. After the mixture was sonicated for another 30 min and stirred at room temperature for 12 h, excess water was evaporated at 333 K with stirring. The solid was then dried at 333 K for 8 h. The as-prepared FeK/rGO catalysts are denoted as FeK0.5/rGO, FeK1/rGO, FeK1.5/rGO, and FeK2/rGO, in which the numbers stand for the nominal K contents of 0.50, 1.0, 1.5, and 2.0 wt %, respectively.

Catalyst Characterization. The multipoint Brunauer−Emmett−Teller surface area (S_{BET}) was measured by N_2 physisorption at 77 K on a Micromeritics TriStar3000 apparatus. Prior to the measurement, the catalyst was pretreated at 473 K under flowing N_2 for 8 h. The Fe and K loadings were determined by inductively coupled plasma atomic emission spectroscopy (ICP-AES; Thermo Elemental IRIS Intrepid).

Conventional powder X-ray diffraction (XRD) patterns were acquired on a Bruker AXS D8 Advance X-ray diffractometer using Ni-filtered Cu Kα radiation (λ = 0.15418 nm). The tube voltage was 40 kV, and the current was 40 mA. The 2θ angles were scanned from 20 to 80° at 1° min^{-1} with a step of 0.01°. The synchrotron radiation XRD patterns were collected on a Huber5021 six-circle diffractometer at the BL14B1 beamline at the Shanghai Synchrotron Radiation Facility (SSRF). The catalyst was protected by PEG200 to avoid oxidation during data acquisition. The X-ray beam wavelength (λ) was set to 0.12398 nm by a Si(111) channel-cut monochromator. The diffractogram was recorded from 30 to 60° at 1° min^{-1} with a step of 0.02°. The synchrotron radiation XRD pattern was then converted to 2θ values corresponding to Cu Kα radiation to aid in a straightforward comparison.

X-ray absorption near-edge structure (XANES) at the Fe K edge was recorded in the transmission mode at room temperature on the 1W1B beamline at the Beijing Synchrotron Radiation Facility (BSRF). The typical electron beam energy was 2.5 GeV, and the current was 200 mA. The beamline was equipped with a Si(111) double crystal monochromator, and the ionization chambers were used to detect the incident and transmitted beams.

Transmission electron microscopy (TEM) images were taken on a JEOL JEM2011 microscope operated at 200 kV. The sample was ground in an agate mortar, sonicated in anhydrous ethanol, dripped onto a carbon film coated copper grid, and dried in air. Particle size distribution (PSD) histograms were constructed by randomly measuring at least 300 NPs.

Fourier transform infrared spectroscopy (FTIR) was conducted on a Nicolet Nexus 470 IR spectrometer. The sample was finely ground, evenly mixed with KBr, and pelletized. The spectral resolution was 4 cm^{-1}, and 32 scans

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Table 1. Basic Physicochemical Properties of the Fe/rGO and FeK/rGO Catalysts

catalyst	K loading[a] (wt %)	Fe loading[a] (wt %)	S_{BET}[b] (m^2 g^{-1})	V_{pore}[b] (cm^3 g^{-1})	d_{pore}[b] (nm)	I_D/I_G[c]
Fe/rGO	0.02	18.2	122	0.33	6.1	1.26
FeK0.5/rGO	0.52	18.1	126	0.34	6.0	1.31
FeK1/rGO	1.03	17.8	135	0.36	5.8	1.31
FeK1.5/rGO	1.51	17.5	138	0.37	5.7	1.29
FeK2/rGO	2.01	17.2	145	0.39	5.5	1.26

[a]Determined by ICP-AES. [b]Determined by N_2 physisorption. [c]Raman intensity ratio between the D and G bands.

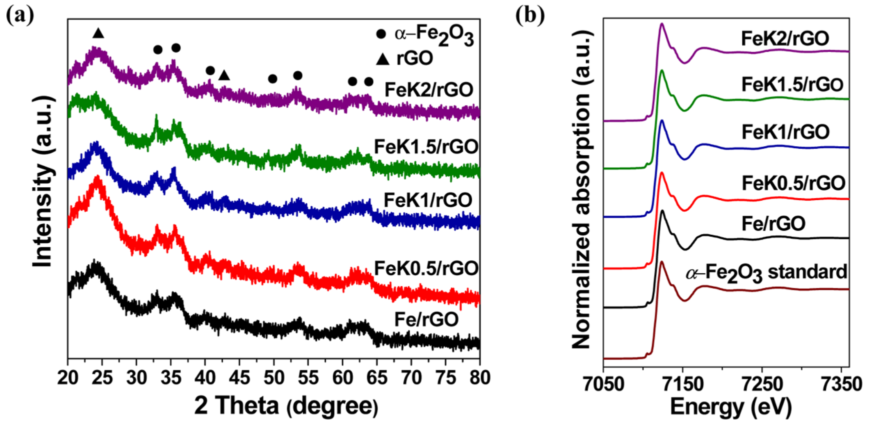

Figure 1. (a) XRD patterns and (b) Fe K-edge XANES spectra of the Fe/rGO and FeK/rGO catalysts. The Fe K-edge XANES spectrum of the α-Fe_2O_3 standard is included in (b) as a reference.

were recorded for each spectrum. The Raman spectrum was recorded at room temperature on a Horiba Jobin Yvon XploRA Raman spectrometer using a 12.5 mW laser source at an excitation wavelength of 532 nm. The spectral resolution was 1.8 cm^{-1}. X-ray photoelectron spectroscopy (XPS) was performed on a Kratos Axis Ultra Dld spectrometer with Al Kα radiation ($h\nu$ = 1486.6 eV) as the excitation source. The pass energies were 160 eV for survey spectra and 40 eV for selected regions.

Temperature-programmed desorption (TPD) of CO_2, CO, or H_2 was performed on a Micromeritics 2750 chemisorption system. Helium was used as the carrier gas in CO_2- and CO-TPD, and Ar was used in H_2-TPD. The weighed catalyst (~100 mg) loaded in a U-shaped quartz tube was reduced at 723 K for 16 h in 5 vol % H_2/Ar (50 mL min^{-1}). After the sample was cooled to 298 K, CO_2, CO, or H_2 pulses were injected until the eluted peak did not change in intensity, as monitored by a thermal conductivity detector (TCD). The catalyst was then purged by carrier gas (25 mL min^{-1}) for at least 30 min to remove the gaseous and physisorbed adsorbate until the baseline was restored. The desorption curve was acquired by heating the catalyst from 300 to 723 K at 10 K min^{-1}. The amount of the desorbed adsorbate was monitored by TCD. Comparison of the area of the desorption peak with that calibrated by a 100 μL capacity loop using the corresponding adsorbate allowed quantification. As the baseline of the H_2-TPD profile was not flat, the desorption curve was corrected by subtracting a linear background before integrating the peak area.

^{57}Fe Mössbauer spectra were recorded in the constant acceleration transmission mode on a Wissel 1550 electromechanical spectrometer (Wissenschaftliche Elektronik GmbH) using a ^{57}Co in Pd matrix irradiation source. The velocity was calibrated by a 25 μm thick α-Fe foil. The isomer shift (IS) was referenced to α-Fe at room temperature. The spectrum was fitted using a least-squares fitting routine that modeled the spectrum as appropriate superpositions of quadruple doublets and magnetic sextets with Lorentzian line shape and constraints in peak width and intensity using the MossWinn 3.0i program. The phase compositions were derived from the areas of the absorption peaks with the assumption of the same recoil-free factor (the probability of absorption of the γ photons) for all kinds of iron nuclei in the catalyst.

Catalytic Testing and Product Analysis. The Fe/rGO and FeK/rGO catalysts were pelletized, crushed, and sieved to 60−80 mesh. Prior to FTO, 100 mg of catalyst was diluted with 400 mg of quartz powder (60−80 mesh) and reduced on site in flowing 5 vol % H_2/Ar (50 mL min^{-1}) at 723 K for 16 h with a heating rate of 2 K min^{-1}. Catalytic testing was performed under industrially relevant operation conditions of 613 K, 20 bar, and a H_2/CO/N_2 ratio of 48/48/4 by volume[1] in a tubular fixed-bed reactor with an inner diameter of 10 mm. N_2 was used as the internal standard during gas-phase product analysis. The FTO product analysis was conducted online using two gas chromatographs fixed with two high-temperature, high-pressure Valco six-port valves. H_2, N_2, CO, CH_4, and CO_2 were analyzed by a GC122 gas chromatograph equipped with a 2 m long TDX-01 packed stainless steel column connected to a TCD. The hydrocarbons were analyzed by a GC9160 gas chromatograph equipped with a PONA capillary column (50 m × 0.25 mm × 0.50 μm) connected to a flame ionization detector (FID). In addition, the C_1−C_4 olefins and paraffins in the gas phase were analyzed by the GC122 gas chromatograph equipped with a PoraPlot Q capillary column (12.5 m × 0.53 mm × 20 μm) connected to a FID. The catalytic activity was expressed as FTY, and the productivity of lower olefins was represented by FTY_{ole}. The hydrocarbon selectivities were calculated on a carbon basis, with the exception of CO_2. The carbon balance of the FTO products in all catalytic runs was better than 92%.

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■ RESULTS AND DISCUSSION

Basic Physicochemical Properties. The S_{BET} values of the Fe/rGO and FeK/rGO catalysts are given in Table 1. Fe/rGO had an S_{BET} value of 122 m^2 g^{-1}. With the modification of K, the S_{BET} increased from 126 to 145 m^2 g^{-1} with a nominal K content from 0.50 to 2.0 wt %, which may be due to the intercalation of K_2CO_3 into the graphene layers under ultrasonication. Table 1 also summarizes the practical loadings of Fe and K on these catalysts. For Fe/rGO, the Fe loading was 18.2 wt %. Interestingly, about 0.02 wt % of K was detected on this catalyst, which may be carried over from the GO precursor using $KMnO_4$ as the oxidant during its synthesis via the Hummers method. No other impurities were detectable by ICP-AES and XRF. For the FeK/rGO catalysts, the practical Fe loadings were in the range of 17.2−18.1 wt %, and the practical K loadings were in good agreement with the nominal values.

Phase and Morphology. Figure 1a shows the XRD patterns of the Fe/rGO and FeK/rGO catalysts. The diffraction peaks at 2θ values of 33.1, 35.6, 40.8, 49.5, 54.1, 62.5, and 64.0° marked with circles are readily assignable to hematite (α-Fe_2O_3, JCPDS 33-0664) on these catalysts. Interestingly, no diffraction peaks due to metallic iron were identified, which may be attributed to the high stability constant of $Fe(acac)_3$ ($\log \beta_3$ = 26.7) that greatly hinders the release and subsequent reduction of ferric ions by hydrazine hydrate to metallic iron. This assignment is corroborated by the XANES spectra in Figure 1b, which show that the Fe K-edge photon energies (7124.1 eV) of these catalysts are identical with that of the α-Fe_2O_3 standard.[22] The crystallite size of α-Fe_2O_3 on Fe/rGO derived from the full width at half-maximum (fwhm) of the most intensive (104) reflection at a 2θ value of 35.6° and the Scherrer equation is ca. 5.7 nm. There was no obvious change in the crystallite size of α-Fe_2O_3 after the modification of K. The diffraction peaks at 2θ values of 24.3 and 42.9° marked with triangles in Figure 1a are due to the (002) and (100) reflections of rGO, respectively, stemming from the formation of the "regraphitized" carbon regions because of the van der Waals attractive interactions.[23,24] Because of the low content and/or high dispersion, there were no discernible diffraction peaks related to K_2CO_3 for the K-promoted catalysts.

Figure 2 presents the TEM images and corresponding PSD histograms with Gaussian analysis fittings of the Fe/rGO and FeK/rGO catalysts. The sheetlike morphology with typical wrinkle of graphene was clearly observed on these catalysts. The graphene sheets were decorated randomly by well-dispersed near-spherical α-Fe_2O_3 NPs of about 5.8−6.0 nm in diameter, which were essentially unchanged after the modification of K. The particle size measured by TEM is in accordance with that estimated by XRD, signifying that the α-Fe_2O_3 NPs are single crystalline.

Surface Properties of rGO in the Catalysts. Figure 3 illustrates the FTIR spectra of the Fe/rGO and FeK/rGO catalysts. The FTIR spectrum of GO recorded as a reference gives the stretching vibrations of O−H (3400 cm^{-1}), C═O (1722 cm^{-1}), aromatic C═C (1620 cm^{-1}), carboxyl O═C−O (1356 cm^{-1}), epoxyl C−O (1220 cm^{-1}), and alkoxyl C−O (1050 cm^{-1}).[25] On the Fe/rGO and FeK/rGO catalysts, the characteristic vibration peaks of the oxygen-containing groups of GO were drastically diminished, whereas two broad peaks emerged at 1550 and 1175 cm^{-1} ascribable to aromatic C═C and C−O stretching vibrations of rGO, respectively.[25] The C−O stretching vibration arises from the defects associated with

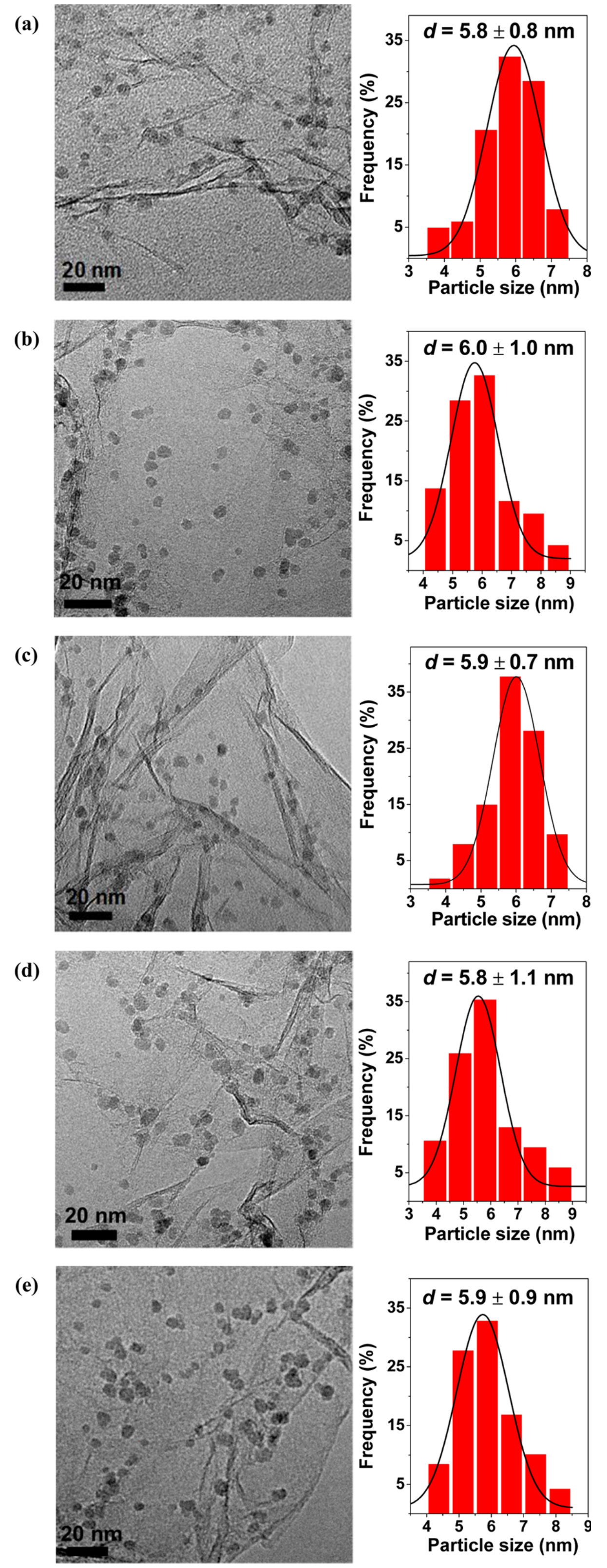

Figure 2. TEM images and PSD histograms with Gaussian analysis fittings of the α-Fe_2O_3 NPs of the (a) Fe/rGO, (b) FeK0.5/rGO, (c) FeK1/rGO, (d) FeK1.5/rGO, and (e) FeK2/rGO catalysts.

residual oxygen in graphene that cannot be completely removed by chemical reduction methods.[26] In brief, the FTIR spectra confirm that GO was reduced to rGO by hydrazine hydrate; however, rGO still contained a small amount of

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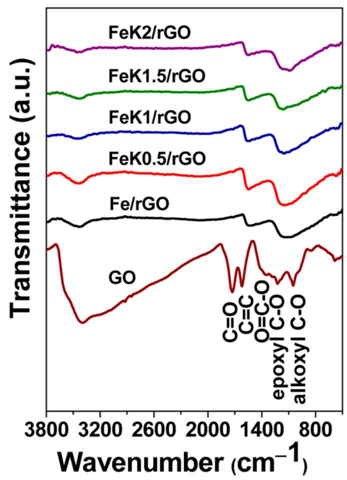

Figure 3. FTIR spectra of GO and the Fe/rGO and FeK/rGO catalysts.

oxygen-containing groups. The absence of the vibration bands at ca. 1350 and 1450 cm^{-1} in Figure 3 characteristic of the carbonate group[27] is consistent with the low contents of K_2CO_3.

Raman spectroscopy is well-suited to identify the defects and the degree of disorder in graphene. As shown in Figure 4a, the

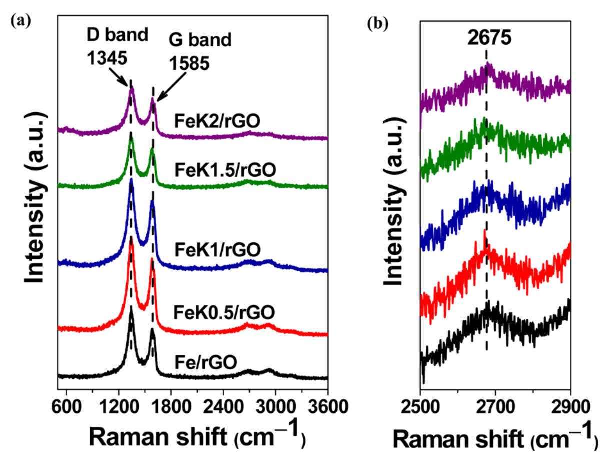

Figure 4. (a) Raman spectra of the Fe/rGO and FeK/rGO catalysts. (b) Corresponding second-order region of the Raman spectra.

Raman spectra of the Fe/rGO and FeK/rGO catalysts display two distinct peaks due to the D and G bands at ca. 1345 and 1585 cm^{-1}, respectively.[28] The G band arises from the vibration of the sp^2-hybridized carbon atoms, whereas the D band comes from the structural disorder at the defects. The D band to G band intensity ratio (I_D/I_G) is usually employed as a measure of the disorder in carbon materials.[29] As also shown in Table 1, the I_D/I_G ratio of Fe/rGO was 1.26. There was no significant change in the I_D/I_G ratio over the K-promoted catalysts, and also no discernible Raman shift for these bands, manifesting the similar nature of rGO in these catalysts. Figure 4b shows that for these catalysts the second-order Raman feature is at around 2675 cm^{-1}. The position of the second-order peak closely depends on the thickness of the graphene layers. It has been reported that the graphene nanosheets of more than five layers bear broad second-order peaks shifting significantly to positions higher than 2700 cm^{-1}.[30,31] It is thus estimated that the thickness of rGO in the Fe/rGO and FeK/rGO catalysts is less than five layers.

XPS was employed to examine the surface species on the Fe/rGO catalyst and the FeK1/rGO catalyst as representative of the K-promoted catalysts. Figure 5a presents the survey scan

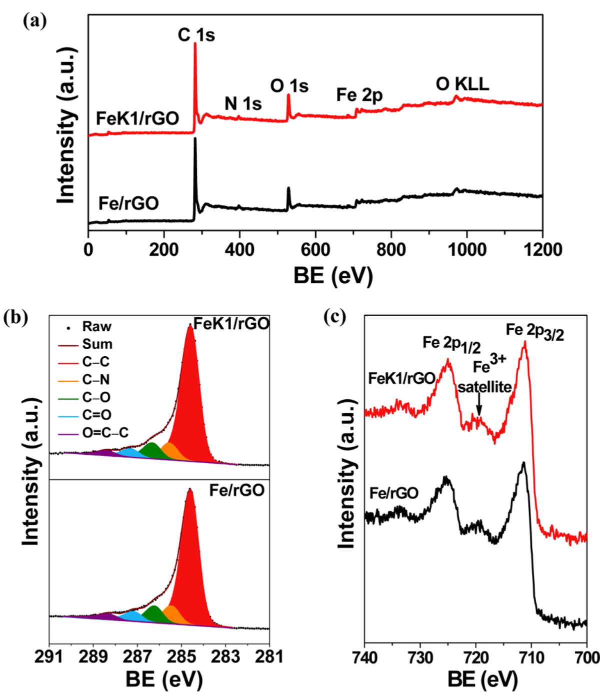

Figure 5. XPS spectra of the Fe/rGO and FeK1/rGO catalysts: (a) survey spectra; (b) C 1s; (c) Fe 2p.

XPS spectra of the catalysts, which reveal that C, O, and Fe are the primary surface species. Both catalysts also gave a weak N 1s peak with similar intensities due to the incorporation of N into rGO reduced by hydrazine hydrate,[21] which is in line with the results of elemental analysis. The signal of K could not be discerned in the spectrum of FeK1/rGO owing to the low K content. Figure 5b displays the C 1s spectra of the Fe/rGO and FeK1/rGO catalysts, which were deconvoluted into five peaks that correspond to carbon atoms in different chemical environments. The peaks with binding energies (BEs) of ca. 284.6, 285.6, 286.4, 287.4, and 288.6 eV are assigned to nonoxygenated ring C, C in C−N, C in C−O, carbonyl C (C═O), and carboxyl C (O═C−O), respectively.[21] In comparison to the survey scan (Figure S1 in the Supporting Information) and C 1s (Figure S2 in the Supporting Information) spectra of GO, both the O 1s peak and the C 1s peaks of the oxygen-containing groups on the Fe/rGO and FeK1/rGO catalysts are drastically attenuated due to the reduction of GO by hydrazine hydrate, which is in agreement with the FTIR results. Although the nonoxygenated ring C dominates the C 1s spectra of the catalysts, the features from heteroatom-containing groups are not negligible. The presence of the heteroatom-containing groups and the defects on rGO disclosed by FTIR, Raman, and XPS is beneficial to the homogeneous distribution of the supported iron NPs.[14] Figure 5c shows the Fe 2p spectra of the Fe/rGO and FeK1/rGO catalysts. The peaks with BEs of 711.1 and 724.8 eV for the Fe $2p_{3/2}$ and $2p_{1/2}$ levels of Fe^{3+}, respectively, coupled with the satellite peak at ∼719.0 eV, clearly verify the formation of Fe_2O_3.[32]

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Surface Basicity and Interaction with CO and H_2. The promotion effect of K on the basicity of the rGO-supported iron catalysts was determined by CO_2-TPD; the profiles are shown in Figure 6a. Fe/rGO displayed a strong peak at about 352 K for weakly chemisorbed CO_2 and a shoulder peak at about 450 K for strongly chemisorbed CO_2.[33] The main peak shifted to 362 K at the nominal K content of 0.50 wt % and retained there at higher K contents, indicating that the strength of the basic sites on the K-promoted catalysts was not affected by K, at least within the nominal content of 2.0 wt %. However, the amount of CO_2 adsorbed increased almost linearly from 204 to 293 μmol g_{cat}^{-1} with the nominal K content from 0 to 2.0 wt % (Figure 6d). The CO_2-TPD characterization confirms that K could increase the amount of basic sites on the rGO-supported iron catalysts, similar to the role of Mn in the iron−manganese catalysts.[34]

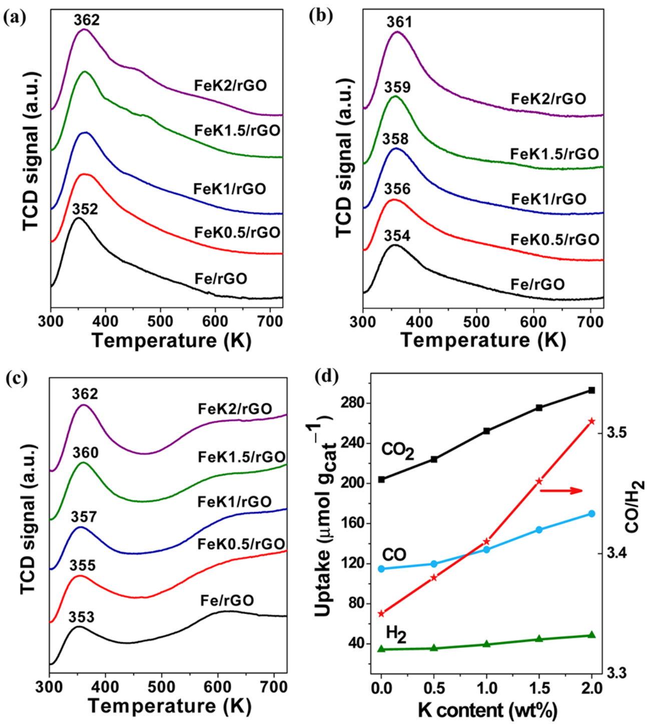

Figure 6. TPD profiles of (a) CO_2, (b) CO, and (c) H_2 of the Fe/rGO and FeK/rGO catalysts. (d) Adsorption capacities of the gases and the ratio of the adsorbed CO to H_2 (CO/H_2) against the K content.

Figure 6b shows the CO-TPD profiles of the Fe/rGO and FeK/rGO catalysts. The desorption temperature of CO increased from 354 to 361 K with a nominal K content from 0 to 2.0 wt %, indicating that increasing the K content could enhance the interaction between iron and CO. Meanwhile, the amount of adsorbed CO increased from 115 to 170 μmol g_{cat}^{-1} (Figure 6d), showing that K enhanced the adsorption of CO not only in the strength but also in the capacity. The effect of K on the adsorption of CO observed here is consistent with previous observations,[16,35,36] which is attributed to the improved basicity of the catalyst by the modification of K (Figure 6a) that facilitates electron donation from the iron to CO.[37]

Figure 6c presents the H_2-TPD profiles of the Fe/rGO and FeK/rGO catalysts. All profiles show a strong desorption peak at above 350 K due to weakly chemisorbed H_2 and a relatively small and broad desorption peak at about 600 K due to strongly chemisorbed H_2.[36] Interestingly, a gradual increase in the desorption temperature of the main peak with the nominal K content from 0 to 2.0 wt % was observed, and the amount of adsorbed H_2 increased synchronously from 34.3 to 48.4 μmol g_{cat}^{-1}. This phenomenon is somewhat unexpected, as the adsorption of H_2 is favored on an electron-deficient metal.[37] The improved surface basicity by K should have suppressed the adsorption of H_2 on the FeK/rGO catalysts.[34,38] Therefore, the electronic effect of K imposed on iron is unlikely to be responsible for the enhanced adsorption of H_2 observed here. Instead, theoretical calculations pointed out that the binding energy of hydrogen atom on graphene could be increased by up to 82% due to the presence of K.[39] It seems that the effect of the K-decorated rGO on enhancing the adsorption of H_2 was so pronounced that a net increase in the amount of adsorbed H_2 resulted in the present case.

FTO Results. Table 2 summarizes the catalytic activities and product distributions over the Fe/rGO and FeK/rGO catalysts under the industrially relevant conditions of 613 K, 20 bar, and a H_2/CO ratio of 1. On Fe/rGO, the FTY was 333 μmol_{CO} g_{Fe}^{-1} s^{-1}. After a modification of only 0.50 wt % of K, the FTY dramatically increased to 556 μmol_{CO} g_{Fe}^{-1} s^{-1} and then maximized at 646 μmol_{CO} g_{Fe}^{-1} s^{-1} over FeK1/rGO, which is higher than the activities reported so far on other supported iron catalysts for FTO.[1,6,9] Further increasing the K content to 1.5 wt % led to an abrupt drop in the activity, and the FTY over FeK2/rGO was even lower than that of Fe/rGO. The effect of K on the activity of the rGO-supported iron catalysts is similar to that of the K-modified Mn/FeN/CNT catalysts, for which a volcanic evolution of the activity with respect to the K content was observed.[40]

As to the hydrocarbon distributions over the Fe/rGO and FeK/rGO catalysts, Table 2 shows that, for Fe/rGO, the selectivity to CH_4 was 48%, while those to C_2−C_4 olefins and paraffins were 31% and 20%, respectively, with the O/P ratio being 1.5. For comparison, the Fe/AC catalyst prepared by the same method as for Fe/rGO but using activated carbon (AC,

Table 2. FTO Results over the Fe/rGO and FeK/rGO Catalysts[a]

catalyst	FTY (μmol_{CO} g_{Fe}^{-1} s^{-1})	space velocity (L h^{-1} g^{-1})	hydrocarbon selectivity (wt %)[b]				O/P	FTY_{ole} (μmol_{CO} g_{Fe}^{-1} s^{-1})
			CH_4	C_2−C_4 olefins	C_2−C_4 paraffins	C_{5+}		
Fe/rGO	333	36	48	31	20	0.3	1.5	105
FeK0.5/rGO	556	60	31	51	14	3.3	3.7	286
FeK1/rGO	646	72	26	62	7.9	4.4	7.8	399
FeK1.5/rGO	271	30	22	67	6.8	5.0	9.8	181
FeK2/rGO	220	24	20	68	6.2	6.7	11	149

[a]Reaction conditions: 0.10 g of catalyst, 613 K, 20 bar, H_2/CO = 1, and TOS = 24 h. Data are reported at CO conversion levels between 58 and 64%; about 49−52% of CO is converted to CO_2. [b]The hydrocarbon selectivities are normalized, with the exception of CO_2.

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Vulcan X72) as the support gave an O/P ratio of only 0.2, manifesting the advantage of the 2D structure of rGO in FTO. When the nominal K content was increased to 2.0 wt %, the selectivities to the unwanted CH_4 and C_2-C_4 paraffins were favorably reduced to 20% and 6.2%, respectively, whereas those to lower olefins increased to 68%, resulting in an O/P ratio as high as 11. It is noteworthy that the selectivities to lower olefins over FeK1/rGO, FeK1.5/rGO, and FeK2/rGO under the industrially relevant reaction conditions are substantially higher than those over K/Mn/FeN/CNT (43.6%),[40] Fe/NCNTs (46.7%),[16] Fe/NG (49.6%),[12] Fe/α-Al_2O_3 modified by Na and S (53%),[1] and Mn/Fe_3O_4 microspheres (60.1%).[41] Although on hierarchically structured α-Al_2O_3-supported iron catalysts modified with S a high selectivity to lower olefins of 68% was reported, it should be noted that this value was obtained at a very low CO conversion of 0.9%.[6] Therefore, our work clearly demonstrates that the high activity and high selectivity to lower olefins could be achieved simultaneously on a properly promoted, designer-made catalyst. In addition, the selectivity to the C_{5+} hydrocarbons increased insignificantly from 0.3 to 6.7% with a nominal K content from 0 to 2.0 wt %, showing that the modification of K did not give an excessive increase in the chain growth probability. Because the decrease in the activity was more pronounced than the increase in the selectivity to lower olefins at a nominal K content above 1.0 wt %, the highest productivity of lower olefins occurred on FeK1/rGO with an FTY_{ole} value of 399 $\mu mol_{CO}\ g_{Fe}^{-1}\ s^{-1}$.

In view of the promising FTY_{ole} over FeK1/rGO, this catalyst was subjected to a 120 h test to examine the stability of the K-promoted rGO-supported iron catalyst in FTO. During the stability test, the space velocity was adjusted to 36 L h^{-1} g^{-1} to give rise to an initial CO conversion of 77%, close to the conversion levels on Fe/α-Al_2O_3 modified with Na and S during stability tests.[1] Figure 7 shows that the FTY increased

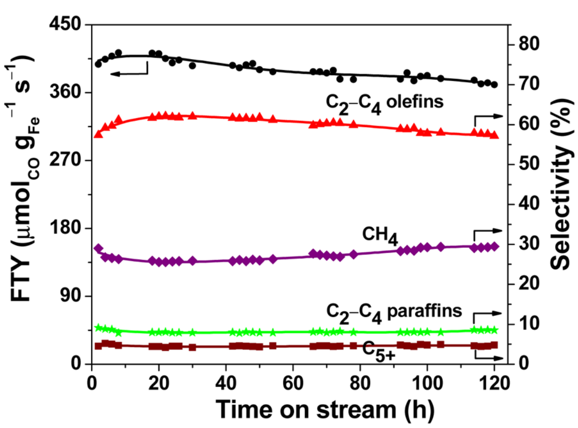

Figure 7. FTO results over the FeK1/rGO catalyst during 120 h on stream at 613 K, 20 bar, a H_2/CO ratio of 1, and a space velocity of 36 L h^{-1} g^{-1}.

slightly with time on stream and reached a maximum within 20 h. Then, the FTY decreased slowly and retained a value of 371 $\mu mol_{CO}\ g_{Fe}^{-1}\ s^{-1}$ after 120 h on stream. As to the hydrocarbon distributions, the selectivities to CH_4 and C_2-C_4 paraffins decreased, whereas those to lower olefins increased during the initial 20 h on stream. Afterward, the selectivity to CH_4 increased slightly to 30% at 120 h of reaction. A marginal increase was also observed in the selectivity to C_2-C_4 paraffins. At the same time, the selectivity to lower olefins declined slightly from 62% to 57%. Although FeK1/rGO suffered from deactivation under the industrially relevant conditions, we found that its degree of deactivation is less than that of 6 wt % Fe/α-Al_2O_3 modified with Na and S and is comparable to that of 12 wt % Fe/α-Al_2O_3 modified with Na and S on comparison over the same time span of 64 h.[1] In addition, FeK1/rGO is far more stable than other iron-based FTO catalysts with the stability being examined. For instance, the CO conversion dropped from about 80% to 55% after 50 h on stream at 613 K, 25 bar, and a H_2/CO ratio of 1 over 20Fe/N-CNT with an iron loading similar to that for FeK1/rGO.[9] Zhou et al. observed an even more drastic activity loss by 60% after only 20 h on stream over the hierarchically structured α-Al_2O_3-supported iron at 623 K, 1 bar, and a H_2/CO ratio of 1.[6] The superior stability of FeK1/rGO in FTO may originate from the anchoring of the iron NPs by the oxygen-containing groups and defects on rGO and the excellent thermal conductivity of rGO, which effectively impede the sintering of the active sites at high temperature.

Active Phase in FTO. The phase composition of the Fe/rGO and FeK/rGO catalysts after 24 h on stream in FTO was examined by synchrotron radiation XRD, which is more sensitive to the weak diffraction peaks of iron carbides than conventional XRD.[42] Figure 8 shows that the predominant iron

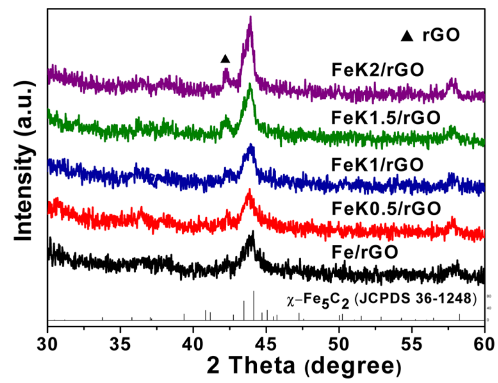

Figure 8. Synchrotron radiation XRD patterns of the Fe/rGO and FeK/rGO catalysts after 24 h on stream in FTO.

phase in these catalysts was Hägg carbide (χ-Fe_5C_2, JCPDS 36-1248). No α-Fe, magnetite (Fe_3O_4), or iron carbides other than χ-Fe_5C_2 were detected by XRD. Hägg carbide is generally acknowledged as the active phase for FTS at around 543 K.[43−50] The present work, as well as the works by de Jong and co-workers[1,51] and Muhler and co-workers,[9] evidence that the active phase for FTO remains Hägg carbide despite the much higher reaction temperature. It is possible that the low H_2/CO ratio adopted in FTO canceled out the destabilization effect of high temperature on Hägg carbide.[52] In addition, the diffraction peaks of Hägg carbide were intensified with the K content. On the basis of the most intense (510) reflection of Hägg carbide at 2θ of 44.1° and the Scherrer equation, the average crystallites size of Hägg carbide were estimated to be ca. 10.5, 10.9, 11.3, 15.8, and 16.1 nm at nominal K contents of 0, 0.50, 1.0, 1.5, and 2.0 wt %, respectively.

The TEM images and PSD histograms of the Fe/rGO and FeK/rGO catalysts after 24 h on stream in FTO are shown in Figure 9. It is observed that the particle sizes of Hägg carbide were ca. 11.1, 11.6, 11.4, 16.0, and 17.0 nm at nominal K contents of 0, 0.50, 1.0, 1.5, and 2.0 wt %, respectively, which are in compliance with the XRD results, indicating that the

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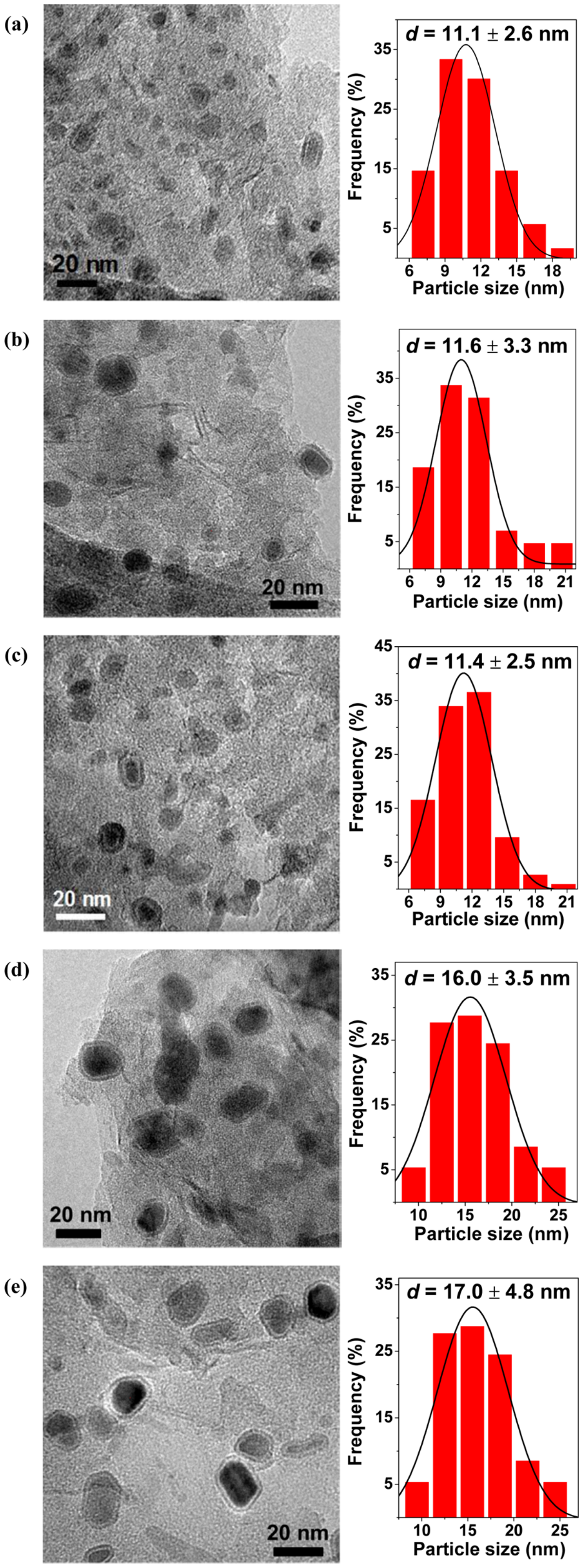

Figure 9. TEM images and PSD histograms with Gaussian analysis fittings of (a) Fe/rGO, (b) FeK0.5/rGO, (c) FeK1/rGO, (d) FeK1.5/rGO, and (e) FeK2/rGO catalysts after 24 h on stream in FTO.

Hägg carbide NPs on rGO are also single crystalline. The relatively small sizes of the Hägg carbide NPs after 24 h on stream at 613 K are attributed to the presence of abundant oxygen-containing groups and defects on rGO as the anchoring sites that effectively inhibit their aggregation.[14] A close inspection of the TEM images reveals that some of the Hägg carbide NPs (especially the larger ones) were covered with a thin carbon layer due to carbon deposition during FTO, for the formation of carbon deposits is favored at high temperatures and low H_2/CO ratios.[53,54] Figure S3 in the Supporting Information shows a TEM image of the FeK1/rGO catalyst after 120 h on stream in FTO. It is shown that the thickness of the carbon layer increased to ~2.7 nm as compared to ~2.4 nm for this catalyst after 24 h on stream in FTO (Figure 9c), which may be responsible for the slight deactivation of the FeK1/rGO catalyst during the 120 h stability test (Figure 7).

Mössbauer spectroscopy is a powerful tool for identifying and quantifying iron phases in iron-based catalysts. Figure 10 presents the ^{57}Fe Mössbauer spectra of the Fe/rGO and FeK/rGO catalysts after 24 h on stream in FTO and the deconvoluted subspectra. These spectra could be best fitted by five sextets and one doublet, and the corresponding fitting parameters are given in Table 3. There is a superparamagnetic (spm) doublet with an IS value of ca. 0.34 mm s^{-1} and a quadrupole splitting (QS) of ca. 0.98 mm s^{-1} assignable to Fe(II) or Fe(III) species due to the presence of some poorly crystallized iron oxide.[42] The magnetite phase is confirmed by two sextets with IS and hyperfine magnetic field (H) values of 0.55−0.57 mm s^{-1} and 43.2−44.3 T and of 0.29−0.39 mm s^{-1} and 47.4−48.6 T corresponding to the tetrahedral (A site) and octahedral sites (B site) of magnetite, respectively.[55,56] Note that the magnetite phase was invisible by synchrotron radiation XRD, suggesting its small crystallite size was below the detection limit of XRD. The Hägg carbide phase is validated by three sextets with H values of 16.3−16.6, 22.0−22.4, and 10.0 T.[1,57] Table 3 compiles the contents of the iron-containing phases in these catalysts after 24 h on stream in FTO. When the nominal K content was increased from 0 to 2.0 wt %, the contents of the spm Fe^{2+}/Fe^{3+} species and magnetite decreased monotonically, whereas that of Hägg carbide increased favorably from 33.8% to 68.5%, which agrees well with the intensification of the diffraction peaks of Hägg carbide with the K content in Figure 8.

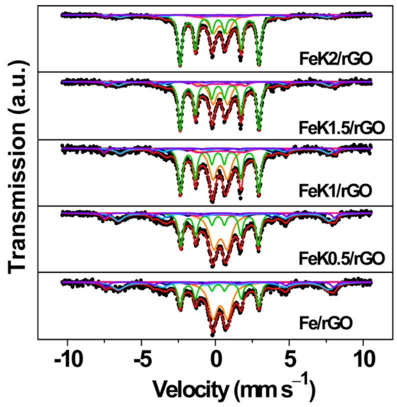

Figure 10. ^{57}Fe Mössbauer spectra of the Fe/rGO and FeK/rGO catalysts after 24 h on stream in FTO.

Structure−Performance Correlation. As mentioned above, Hägg carbide has been identified as the active phase for FTO. Figure 10 and Table 3 reveal that K effectively promoted the carbidation of the rGO-supported iron catalysts to Hägg carbide and/or stabilized this phase under the harsh

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Table 3. ^{57}Fe Mössbauer Parameters of the Fe/rGO and FeK/rGO Catalysts after 24 h on Stream in FTO[a]

catalyst	IS (mm s^{-1})	QS (mm s^{-1})	H (T)	Γ (mm s^{-1})	phase ascription	A (%)
Fe/rGO	0.33	0.98		0.74	Fe(II)/Fe(III)	36.3
	0.24	0.08	16.5	0.34	Fe_5C_2 (A)	24.9
	0.24	0.1	22.4	0.58	Fe_5C_2 (B)	4.9
	0.21	0.02	10.0	0.88	Fe_5C_2 (C)	4.0
	0.55		44.1	0.10	Fe_3O_4 (A)	22.6
	0.29		48.0	0.46	Fe_3O_4 (B)	7.3
FeK0.5/rGO	0.35	0.92		0.76	Fe(II)/Fe(III)	33.6
	0.24	0.09	16.3	0.34	Fe_5C_2 (A)	30.9
	0.24	0.1	22.0	0.58	Fe_5C_2 (B)	6.8
	0.34	0.02	10.0	0.86	Fe_5C_2 (C)	5.2
	0.57		44.3	0.84	Fe_3O_4 (A)	18.1
	0.29		48.4	0.39	Fe_3O_4 (B)	5.4
FeK1/rGO	0.33	0.98		0.65	Fe(II)/Fe(III)	27.5
	0.23	0.1	16.5	0.33	Fe_5C_2 (A)	42.7
	0.24	0.1	22.0	0.58	Fe_5C_2 (B)	6.3
	0.34	0.03	10.0	0.88	Fe_5C_2 (C)	8.3
	0.61		43.8	0.77	Fe_3O_4 (A)	11.6
	0.31		48.6	0.29	Fe_3O_4 (B)	3.6
FeK1.5/rGO	0.34	0.10		0.60	Fe(II)/Fe(III)	21.6
	0.24	0.09	16.6	0.32	Fe_5C_2 (A)	51.7
	0.28	0.1	22.1	0.47	Fe_5C_2 (B)	4.6
	0.34	0.10	10.0	0.61	Fe_5C_2 (C)	8.1
	0.56		44.2	0.71	Fe_3O_4 (A)	9.5
	0.39		48.0	0.40	Fe_3O_4 (B)	4.5
FeK2/rGO	0.33	0.95		0.67	Fe(II)/Fe(III)	21.5
	0.24	0.09	16.6	0.33	Fe_5C_2 (A)	61.6
	0.24	0.1	22.4	0.31	Fe_5C_2 (B)	1.6
	0.34	0.02	10.0	0.79	Fe_5C_2 (C)	5.3
	0.55		43.3	0.79	Fe_3O_4 (A)	7.4
	0.35		48.3	0.34	Fe_3O_4 (B)	2.6

[a]Definitions: IS, isomer shift (relative to α-Fe); QS, quadrupole splitting; H, hyperfine magnetic field; Γ, fwhm; A, relative spectral area.

FTO reaction conditions. This effect can be related to the improved basicity of the rGO-supported iron catalysts modified with K (Figure 6a), which favors the chemisorption and dissociation of CO by facilitating electron back-donation from iron to CO.[20,37] This interpretation is nicely substantiated by a good linear relationship between the content of Hägg carbide and the adsorption capacity of CO_2 on the Fe/rGO and FeK/rGO catalysts plotted in Figure 11.

Although the content of Hägg carbide increased with the K content, the activity did not vary proportionally with the former. Actually, a volcano-like evolution of the FTY with respect to the K content was observed. Concomitant with the abrupt drop in the FTY from FeK1/rGO to FeK1.5/rGO, both

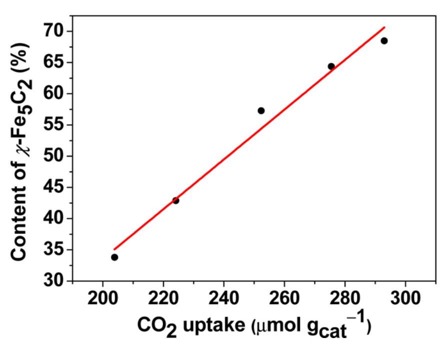

Figure 11. Plot of the surface basicity as a function of the content of Hägg carbide of the Fe/rGO and FeK/rGO catalysts.

XRD (Figure 8) and TEM (Figure 9) characterizations revealed a sudden increase in the size of Hägg carbide from ∼11 to ∼16 nm. The enlarged size of Hägg carbide is expected to reduce the active sites accessible by syngas. In addition, it was reported that the presence of K favors carbon deposition.[17,18] Analogously, Na resulted in a dramatic increase in the amount of the carbon deposits on Fe/α-Al_2O_3 modified with Na and S.[54] On the basis of Figure 9, we measured the thicknesses of the carbon layer of the Fe/rGO and FeK/rGO catalysts as ca. 2.2, 2.4, 2.4, 2.6, and 2.7 nm at nominal K contents of 0, 0.50, 1.0, 1.5, and 2.0 wt %, respectively. Therefore, it is plausible that a combination of the positive role of K in promoting the formation of Hägg carbide and the negative role of excess K in enlarging the size of Hägg carbide and blocking the active phase by K-induced carbon deposition leads to the volcanic evolution of the FTY in the present case.

In the work of de Jong and co-workers, they found that the selectivity to lower olefins remained almost unchanged for iron carbide with a size above 4 nm.[57] In contrast, on the Fe/rGO and FeK/rGO catalysts, the selectivity to lower olefins increased from 31% to 68% with the particle size of Hägg carbide from 10.5 to 16.0 nm, implying that the promotion effect of K far surpassed the particle size effect. According to the Anderson−Schulz−Flory (ASF) plots of the Fe/rGO and FeK/rGO catalysts in Figure 12, the chain growth probability (α) increased from 0.14 to 0.29 with a nominal K content from 0 to 2.0 wt %, which favors chain growth and the termination step via β-hydride abstraction that cannot result in CH_4 production.[1]

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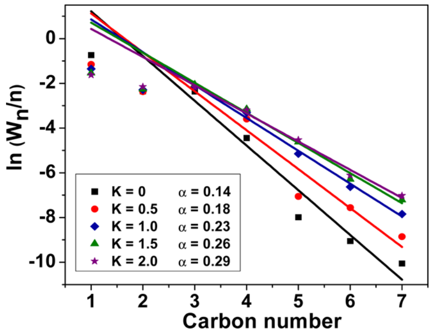

Figure 12. ASF plots of the Fe/rGO and FeK/rGO catalysts.

In other words, the modification of K improves the selectivity to lower olefins by suppressing the production of CH_4, which accounts for the decrease in the selectivity to CH_4 with the K content in Table 2. Furthermore, Figures 6d shows that, with an increase in surface basicity, the amounts of the adsorbed CO and H_2 both increased, with the amplitude of the increment being more remarkable for the former. The improved surface CO/H_2 ratio is adverse to the secondary hydrogenation of lower olefins,[16,20] thus rendering an increase in the selectivity to lower olefins at the expense of the C_2–C_4 paraffins. As a result, the O/P ratio also increased with the K content.

Finally, it is interesting to note that the selectivities to the C_{5+} hydrocarbons on the FeK/rGO catalysts are much lower than those reported in the literature, which were always no less than 12% under the same FTO reaction conditions.[1,51,54] Consistent with this deviation, Figure 12 illustrates that the highest α value over FeK2/rGO is still much smaller than the value of 0.41 over Fe/α-Al_2O_3 modified with Na and S,[1] showing that the modification of K on the rGO-supported iron catalyst did not incur considerable chain growth. It is possible that K in direct contact with rGO instead of Fe does not play a role in enhancing the chain propagation ability of Fe. On the other hand, we propose that the unexpectedly enhanced H_2 adsorption on the K-promoted rGO (Figure 6d) may also contribute to the restrained improvement of the selectivity to the C_{5+} hydrocarbons. It is conceivable that, at an elevated surface concentration of hydrogen, the hydrogenation of the surface carbon species is favored, which competes with the coupling of lower olefins and hence limits the shift of the product spectrum to long-chain hydrocarbons.[20]

CONCLUSIONS

The modification of K imposed profound effects on the physicochemical properties and catalytic performances of the rGO-supported iron catalysts in FTO. With an increment of the nominal K content from 0 to 2.0 wt %, the S_{BET}, surface basicities, adsorption capacities of CO and H_2, and particle sizes of Hägg carbide increased accordingly. A good linear relationship is identified between the surface basicity and the content of Hägg carbide on these catalysts. In FTO, the FTY showed a volcanic evolution with the content of K, which is attributed to the promotion effect of K on the formation of Hägg carbide and the negative effects of K on the enlargement of the size of Hägg carbide and blocking of the active phase by K-induced carbon deposition. The maximum FTY value of 646 $\mu mol_{CO}\ g_{Fe}^{-1}\ s^{-1}$ was obtained on FeK1/rGO. The selectivity to lower olefins and the O/P ratio increased monotonically to 68% and 11, respectively, on FeK2/rGO, which is interpreted as the improved chain-growth probability and surface CO/H_2 ratio due to K modification that effectively suppressed the production of CH_4 and the secondary hydrogenation of lower olefins. This work clearly demonstrates that, using rGO as the support and K as the promoter, it is viable to impart both high activity and high selectivity to lower olefins to iron-based catalyst in the industrially appealing FTO process.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acscatal.5b02024.

Survey scan and C 1s XPS spectra of GO and the TEM image of the FeK1/rGO catalyst after 120 h on stream in FTO (PDF)

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Notes

The authors declare no competing financial interest.

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REFERENCES

(1) Torres Galvis, H. M.; Bitter, J. H.; Khare, C. B.; Ruitenbeek, M.; Dugulan, A. I.; de Jong, K. P. *Science* **2012**, *335*, 835–838.

(2) Diercks, R.; Arndt, J.; Freyer, S.; Geier, R.; Machhammer, O.; Schwartze, J.; Volland, M. *Chem. Eng. Technol.* **2008**, *31*, 631–637.

(3) Torres Galvis, H. M.; de Jong, K. P. *ACS Catal.* **2013**, *3*, 2130–2149.

(4) Schulz, H. *Appl. Catal., A* **1999**, *186*, 3–12.

(5) Cheng, K.; Ordomsky, O. O.; Legras, B.; Virginie, M.; Paul, S.; Wang, Y.; Khodakov, A. Y. *Appl. Catal., A* **2015**, *502*, 204–214.

(6) Zhou, X. P.; Ji, J.; Wang, D.; Duan, X. Z.; Qian, G.; Chen, D.; Zhou, X. G. *Chem. Commun.* **2015**, *51*, 8853–8856.

(7) Sun, B.; Xu, K.; Nguyen, L.; Qiao, M. H.; Tao, F. *ChemCatChem* **2012**, *4*, 1498–1511.

(8) Xiong, H. F.; Jewell, L. L.; Coville, N. J. *ACS Catal.* **2015**, *5*, 2640–2658.

(9) Schulte, H. J.; Graf, B.; Xia, W.; Muhler, M. *ChemCatChem* **2012**, *4*, 350–355.

(10) Xu, J. D.; Zhu, K. T.; Weng, X. F.; Weng, W. Z.; Huang, C. J.; Wan, H. L. *Catal. Today* **2013**, *215*, 86–94.

(11) Wang, D.; Zhou, X. P.; Ji, J.; Duan, X. Z.; Qian, G.; Zhou, X. G.; Chen, D.; Yuan, W. K. *J. Mater. Chem. A* **2015**, *3*, 4560–4567.

(12) Chen, X. Q.; Deng, D. H.; Pan, X. L.; Hu, Y. F.; Bao, X. H. *Chem. Commun.* **2015**, *51*, 217–220.

(13) Cheng, Y.; Fan, Y. Q.; Pei, Y.; Qiao, M. H. *Catal. Sci. Technol.* **2015**, *5*, 3903–3916.

(14) Sun, B.; Jiang, Z.; Fang, D.; Xu, K.; Pei, Y.; Yan, S. R.; Qiao, M. H.; Fan, K. N.; Zong, B. N. *ChemCatChem* **2013**, *5*, 714–719.

(15) Truong-Huu, T.; Chizari, K.; Janowska, I.; Moldovan, M. S.; Ersen, O.; Nguyen, L. D.; Ledoux, M. J.; Pham-Huu, C.; Begin, D. *Catal. Today* **2012**, *189*, 77–82.

(16) Lu, J. Z.; Yang, L. J.; Xu, B. L.; Wu, Q.; Zhang, D.; Yuan, S. J.; Zhai, Y.; Wang, X. Z.; Fan, Y. N.; Hu, Z. *ACS Catal.* **2014**, *4*, 613–621.

(17) Yang, Y.; Xiang, H. W.; Xu, Y. Y.; Bai, L.; Li, Y. W. *Appl. Catal., A* **2004**, *266*, 181–194.

(18) Ma, W. P.; Kugler, E. L.; Dadyburjor, D. B. *Energy Fuels* **2007**, *21*, 1832−1842.
(19) Xu, L. Y.; Wang, Q. X.; Xu, Y. D.; Huang, J. S. *Catal. Lett.* **1995**, *31*, 253−266.
(20) Li, J. F.; Cheng, X. F.; Zhang, C. H.; Yang, Y.; Li, Y. W. *J. Mol. Catal. A: Chem.* **2015**, *396*, 174−180.
(21) Stankovich, S.; Dikin, D. A.; Piner, R. D.; Kohlhaas, K. A.; Kleinhammes, A.; Jia, Y.; Wu, Y.; Nguyen, S. T.; Ruoff, R. S. *Carbon* **2007**, *45*, 1558−1565.
(22) Li, S.; Meitzner, G. D.; Iglesia, E. *J. Phys. Chem. B* **2001**, *105*, 5743−5750.
(23) Scheuermann, G. M.; Rumi, L.; Steurer, P.; Bannwarth, W.; Mülhaupt, R. *J. Am. Chem. Soc.* **2009**, *131*, 8262−8270.
(24) Nie, R. F.; Wang, J. H.; Wang, L. N.; Qin, Y.; Chen, P.; Hou, Z. Y. *Carbon* **2012**, *50*, 586−596.
(25) Chandra, V.; Park, J.; Chun, Y.; Lee, J. W.; Hwang, I.; Kim, K. S. *ACS Nano* **2010**, *4*, 3979−3986.
(26) Li, D.; Müller, M. B.; Gilje, S.; Kaner, R. B.; Wallace, G. G. *Nat. Nanotechnol.* **2008**, *3*, 101−105.
(27) Frost, R. L.; Scholz, R.; López, A. *J. Mol. Struct.* **2015**, *1097*, 1−5.
(28) Machado, B. F.; Serp, P. *Catal. Sci. Technol.* **2012**, *2*, 54−75.
(29) Wang, H. L.; Robinson, J. T.; Li, X. L.; Dai, H. J. *J. Am. Chem. Soc.* **2009**, *131*, 9910−9911.
(30) Ni, Z. H.; Wang, Y. Y.; Yu, T.; Shen, Z. X. *Nano Res.* **2008**, *1*, 273−291.
(31) Siamaki, A. R.; Khder, A. E. R. S.; Abdelsayed, V.; El-Shall, M. S.; Gupton, B. F. *J. Catal.* **2011**, *279*, 1−11.
(32) Lin, T. C.; Seshadri, G.; Kelber, J. A. *Appl. Surf. Sci.* **1997**, *119*, 83−92.
(33) Xu, L. Y.; Wang, Q. X.; Liang, D. B.; Wang, X.; Lin, L. W.; Cui, W.; Xu, Y. D. *Appl. Catal., A* **1998**, *173*, 19−25.
(34) Li, T. Z.; Wang, H. L.; Yang, Y.; Xiang, H. W.; Li, Y. W. *J. Energy Chem.* **2013**, *22*, 624−632.
(35) Zhang, C. H.; Zhao, G. Y.; Liu, K. K.; Yang, Y.; Xiang, H. W.; Li, Y. W. *J. Mol. Catal. A: Chem.* **2010**, *328*, 35−43.
(36) Wan, H. J.; Wu, B. S.; Xiang, H. W.; Li, Y. W. *ACS Catal.* **2012**, *2*, 1877−1883.
(37) Miller, D. G.; Moskovits, M. *J. Phys. Chem.* **1988**, *92*, 6081−6085.
(38) Li, S. Z.; Li, A. W.; Krishnamoorthy, S.; Iglesia, E. *Catal. Lett.* **2001**, *77*, 197−205.
(39) Tapia, A.; Acosta, C.; Medina-Esquivel, R. A.; Canto, G. *Comput. Mater. Sci.* **2011**, *50*, 2427−2432.
(40) Yang, Z. Q.; Pan, X. L.; Wang, J. H.; Bao, X. H. *Catal. Today* **2012**, *186*, 121−127.
(41) Liu, Y.; Chen, J. F.; Bao, J.; Zhang, Y. *ACS Catal.* **2015**, *5*, 3905−3909.
(42) Xu, K.; Sun, B.; Lin, J.; Wen, W.; Pei, Y.; Yan, S. R.; Qiao, M. H.; Zhang, X. X.; Zong, B. N. *Nat. Commun.* **2014**, *5*, 5783.
(43) Bukur, D. B.; Nowicki, L.; Manne, R. K.; Lang, X. S. *J. Catal.* **1995**, *155*, 366−375.
(44) Li, S. Z.; O'Brien, R. J.; Meitzner, G. D.; Hamdeh, H.; Davis, B. H.; Iglesia, E. *Appl. Catal., A* **2001**, *219*, 215−222.
(45) Riedel, T.; Schulz, H.; Schaub, G.; Jun, K. W.; Hwang, J. S.; Lee, K. W. *Top. Catal.* **2003**, *26*, 41−54.
(46) Hayakawa, H.; Tanaka, H.; Fujimoto, K. *Appl. Catal., A* **2006**, *310*, 24−30.
(47) de Smit, E.; Beale, A. M.; Nikitenko, S.; Weckhuysen, B. M. *J. Catal.* **2009**, *262*, 244−256.
(48) Zhang, Q. H.; Kang, J. C.; Wang, Y. *ChemCatChem* **2010**, *2*, 1030−1058.
(49) Sun, B.; Yu, G. B.; Lin, J.; Xu, K.; Pei, Y.; Yan, S. R.; Qiao, M. H.; Fan, K. N.; Zhang, X. X.; Zong, B. N. *Catal. Sci. Technol.* **2012**, *2*, 1625−1629.
(50) Yang, C.; Zhao, H. B.; Hou, Y. L.; Ma, D. *J. Am. Chem. Soc.* **2012**, *134*, 15814−15821.
(51) Torres Galvis, H. M.; Koeken, A. C. J.; Bitter, J. H.; Davidian, T.; Ruitenbeek, M.; Dugulan, A. I.; de Jong, K. P. *Catal. Today* **2013**, *215*, 95−102.
(52) de Smit, E.; Cinquini, F.; Beale, A. M.; Safonova, O. V.; van Beek, W.; Sautet, P.; Weckhuysen, B. M. *J. Am. Chem. Soc.* **2010**, *132*, 14928−14941.
(53) Koeken, A. C. J.; Torres Galvis, H. M.; Davidian, T.; Ruitenbeek, M.; de Jong, K. P. *Angew. Chem., Int. Ed.* **2012**, *51*, 7190−7193.
(54) Torres Galvis, H. M.; Koeken, A. C. J.; Bitter, J. H.; Davidian, T.; Ruitenbeek, M.; Dugulan, A. I.; de Jong, K. P. *J. Catal.* **2013**, *303*, 22−30.
(55) Sirimanothan, N.; Hamdeh, H. H.; Zhang, Y. Q.; Davis, B. H. *Catal. Lett.* **2002**, *82*, 181−191.
(56) Sun, B.; Lin, J.; Xu, K.; Pei, Y.; Yan, S. R.; Qiao, M. H.; Zhang, X. X.; Zong, B. N. *ChemCatChem* **2013**, *5*, 3857−3865.
(57) Torres Galvis, H. M.; Bitter, J. H.; Davidian, T.; Ruitenbeek, M.; Dugulan, A. I.; de Jong, K. P. *J. Am. Chem. Soc.* **2012**, *134*, 16207−16215.

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